SAFEMAN · STUDENT MODULE

SM-WAH

Working at Heights & Fall Arrest

SafeMan™ by Oryx Institute · safeman.oryxinstitute.org

Theory + Harness Practicum · Pass mark 70% · 3 modules

Name _____

Student no. _____ Class _____



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WELCOME

SafeMan Working at Heights & Fall Arrest Protection

SafeMan Working at Heights & Fall Arrest Protection — Theory + Harness Practicum. This SafeMan course is engineered for corporate audit readiness against: GN 156 Reg. 39–45; NTA Unit Std 849; SANS/ISO 10333.

REGULATORY DISCLOSURE — READ FIRST

SafeMan™ is a proprietary inter-organizational safety management and operational competency benchmark developed by Oryx Institute (safeman.oryxinstitute.org). Courses are engineered in alignment with ISO 45001:2018, ISO 14001:2015, ISO 31000, ILO-OSH guidelines, and pre-referenced against Namibia Training Authority (NTA) Unit Standards and Namibian Government Notice 156 of 1997 for corporate audit readiness. SafeMan credentials verify industry competency and attendance between organizations and are non-NQA credit-bearing.

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How to use this module pack

Work through the modules in order. Each module states its focus and the key points you are expected to carry into practice. Diagrams referenced as web assets accompany the online reader at safeman.oryxinstitute.org. After the modules, sit the knowledge check: 10 questions, pass mark 70%, closed book.

This is a practicum course: after passing the knowledge check you must complete the assessor practicum checklist with a qualified assessor before your SafeMan credential is issued. A practical demonstration video slot is reserved in the online reader for every module.

MODULES

Module 1 — Fall physics: fall factor & clearance

Why fall distance, lanyard length and anchor height decide survivability before the harness is even worn.

Key points

Fall factor = fall distance / lanyard length; FF2 (factor 2) doubles the arrest load and is never acceptable.

Clearance = fall distance + lanyard + deceleration distance + body height + safety margin; do the sum before the first step up.

Free fall is limited and shock absorbers tear out to cap arrest force near 6 kN.

Anchors above the D-ring keep fall factor at or near zero.

Diagram: /assets/webp/sm-harness.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

Module 2 — Harness, lanyards & anchor points

Pre-use inspection discipline, 100% tie-off with twin tails, and anchor rating and placement rules.

Key points

Full-body harness inspection: webbing (cuts, burns, chemical stains), stitching, buckles, D-rings, labels — every shift by the user and formally every period.

Twin-tail lanyards give 100% tie-off: one tail always connected while moving.

Anchor: engineered or verified 15 kN per person (per manufacturer standard), above the head where possible.

Scaffold tags: green for complete and inspected, red for incomplete or out of service — respect the tag.

Diagram: /assets/webp/sm-harness.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

Module 3 — Suspension trauma & rescue plan

What happens in a suspended harness after arrest, and why no job goes up without a rescue plan.

Key points

Suspension trauma: legs immobilized, blood pools, fainting and death can follow within minutes.

Relief straps or leg movement buy time; rescue must reach the casualty fast and lower them gently.

No work at height proceeds without a written rescue plan and someone on the ground who can execute it.

Post-arrest: medical assessment even when the worker feels fine.

Diagram: /assets/webp/sm-harness.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

ASSESSMENT

Knowledge check

The closed-book knowledge check draws 10 questions from the bank reproduced in Appendix A. Pass mark 70%. Your SafeMan credential is issued after the knowledge check and the practicum sign-off, and remains valid for 24 months from issue.

Assessor Practicum Checklist — Working at Heights

Pass standard: Inspection, donning and tie-off performed to assessor standard on the training rig.

1	Harness webbing, stitching, buckles and D-rings inspected with faults correctly identified on the seeded example.	
2	Harness donned with correct chest and leg strap tensioning and D-ring placement.	
3	Twin-tail 100% tie-off progression drill completed without unprotected transitions.	
4	Anchor point selected and justified against rating and placement rules.	
5	Rescue plan verbal walk-through delivered for the simulated suspension scenario.	

Assessor name, role/qualification and company are captured on the sign-off screen of the SafeMan platform and bound into the QR-verified certificate record.

APPENDIX A — QUESTION BANK

Knowledge check bank

Q1. Fall factor 2 occurs when:

A. The anchor is level with the D-ring

B. The fall distance is twice the lanyard length

C. Two workers fall

D. The harness slips

Q2. Maximum acceptable arrest force is capped near:

A. 1 kN

B. 6 kN

C. 15 kN

D. 40 kN

Q3. 100% tie-off means:

A. One hook clipped anywhere

B. A twin-tail lanyard keeps one tail connected at all times

C. Tying the rope twice

D. Working only at ground level

Q4. Harness pre-use inspection is done:

A. Monthly by the supplier

B. By the user every shift before use

C. Yearly

D. Only after a fall

Q5. A green scaffold tag certifies:

A. Paint is dry

B. Scaffold complete and inspected for use

C. Maximum two persons

D. Weather approval

Q6. Correct clearance sum includes fall distance, lanyard, deceleration, body height and:

A. Tool weight

B. A safety margin

C. Wind speed

D. Rope age

Q7. Suspension trauma develops because:

A. The harness cuts breathing

B. Immobility pools blood in the legs

C. Heat builds in webbing

D. The lanyard spins

Q8. Anchor points should be:

A. At foot level

B. Above the D-ring where possible and rated for the load

C. Any pipe

D. Shared with crane slings

Q9. After a fall arrest the worker:

A. Returns to work if unhurt

B. Gets medical assessment — always

C. Swaps harnesses and continues

D. Signs an incident note only

Q10. A rescue plan is required:

A. Only for night work

B. Before any work at height proceeds

C. Only above 20 m

D. After the first fall

Answer key

Q1=B Q2=B Q3=B Q4=B Q5=B Q6=B
Q7=B Q8=B Q9=B Q10=B